Alexander J. Dimoff

Max Planck Institute for Astronomy (MPIA), Heidelberg, Germany

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Education

PhD in Astrophysics
Goethe University Frankfurt, Germany *2022–2025*
Dissertation: nuclear astrophysics; spectroscopy, binary stellar evolution, nuclear experiments.
Advisors: Prof. Dr. Camilla J. Hansen and Sen. Lec. Dr. Richard J. Stancliffe (U. Bristol)

M.Sc. in Astronomy
San Diego State University, USA *2019–2021*
Thesis: Apsidal Motion in Eclipsing Binaries.
Advisor: Prof. Dr. Jerome Orosz

B.Sc. in Astrophysics
Pennsylvania State University, USA *2013–2017*
Minors: Physics, German Language & Culture.
Advisor: Evan Pugh Prof. Dr. James Kasting

Research

Postdoctoral Researcher
Max Planck Institute for Astronomy, Heidelberg, Germany *2025–Present*
Star-planet connections through n-capture elements, radiogenic heating in exoplanets.

Doctoral Researcher
Goethe University Frankfurt Institute of Applied Physics, Frankfurt, Germany *2022–2025*
Observational spectroscopy, stellar abundances, binary accretion and evolution, nuclear reaction cross-sections.

Master's Researcher
San Diego State University, San Deigo, USA *2020–2021*
Orbital dynamics of binaries, apsidal motion, stellar structure and evolution.

Post-graduate Research Assistant
Pennsylvania State University, Pennsylvania, USA *2017–2019*
Isotope fractionation in stellar nebulae, modeling equilibrium and kinetic chemistry.

Teaching and Mentoring

Master's Student Co-Supervision
Goethe University Frankfurt, Germany *2024–2025*
Orbital dynamics and exoplanet studies; limb-darkening laws for high-metallicity stars

Teaching Assistant
Goethe University Frankfurt, Germany *2022–2024*
Courses: Astronomy I & II (2 semesters)

Teaching Associate
San Diego State University, San Diego, USA *2020–2021*
Astronomy 109 laboratory (3 semesters)
Astronomy 101 lecture (1 semester)
Flexible Course Design Institute for Graduate Students

Awards and Scholarships

- William and Nhung Booth Scholarship for service and scholastic achievement $6000 prize (2021)
- Ruth and Clifford Smith Graduate Fellowship for scholastic achievement $2000 prize (2020)
- Reginal F. Bueller Endowment Scholarship for community service and engagement $1000 prize (2020)
- William F. Lucas SDAA Endowment Scholarship for scholastic achievement $2000 prize (2020)
- Awona W. Harrington Award for Service $1000 prize (2020)
- NASA Pennsylvania Space Grant, funding to attend workshop (2017)

Observation Proposals

- Nordic Optical Telescope FIES High-resolution spectrograph (67000) – PI, 10 nights 2022
- MPG 2.2m Telescope FEROS High-resolution spectrograph (48000) – PI, 40 nights 2022-2025
- Rozhen Observatory ESpeRo High-resolution spectrograph (30000) – PI, 39 nights 2021-2025
- Moletai Observatory VUES High-resolution spectrograph (30000) – PI, 57 nights 2021-2025
- Ondrejov Observatory OES High-resolution spectrograph (50000) – PI, 46 nights 2021-2025

Skills

Languages: English (native), German B1

Programming: Python, Bash, FORTRAN, Julia, ROOT

Libraries: NumPy, SciPy, Astropy, Pandas, Scikit-Learn, JAX, Numba, Matplotlib

Tools: Git, LaTeX, HPC (MPI/OpenMP, Slurm), ADQL/SQL

Methods: MCMC, Bayesian Inference, Maximum Likelihood, Gaussian Processes, Poisson Statistics, Cross-Correlation, Uncertainty Quantification, Data Visualization

Professional Development

- Max Planck International Research School (IMPRS), PhD student (2022 - 2025)
- Student Representative MPIA, Max Planck Society (2022 – 2024)
- Sustainability Committee Member, MPIA (since 2023)
- Organizer, MPIA Galaxies and Cosmology department retreat (2023)
- Workshops and Schools: ESO Stellar Spectroscopy Workshop (2024), ECR Astronuclear School (2024), Russbach Winter School for Nuclear Astrophysics (2023), Ondrejov Observing School (2022), NPA-X Summer School CERN (2022)

Conferences and Presentations (selected)

- 2025 - S-Process Signatures of Binary Interaction across the Milky Way (AIP, Potsdam, Germany)
- 2025 – Modeling the Progenitors of Low-Mass Post-Accretion Binaries (sirEN Conference, Italy)
- 2024 – Connecting Binary Accretion and Abundances (Nuclear Physics and Astrophysics XI, Germany)
- 2024 – Tracing Carbon from AGB Stars through Binary Accretion (NAOJ, Tokyo)
- 2024 – S-Process Nucleosynthesis in AGB Stars (IAP Seminar, Frankfurt; XIV Torino AGB Workshop, Italy)
- 2023 – s- and i-Process Nucleosynthesis (i-Process Workshop, Cyprus; Russbach Winter School, Austria)
- 2022 – Spectral Observations of s-Process Stars (ChETEC-INFRA, Italy; Ondrejov School, Czech Republic)

Publications

- **Dimoff**, Bergemann, Foley et al. (in prep). No Star, No Life? Radiogenic Heating in Exoplanets and Extending the Habitable Zone in Chemically Enhanced Systems.
- **Dimoff**, Stancliffe, Hansen et al. (2025). Modeling the Progenitors of Low-Mass Post-Accretion Binaries. *Astronomy & Astrophysics*.
- **Dimoff**, Hansen, Stancliffe et al. (2024). S-Process Nucleosynthesis in Chemically Peculiar Binaries. *Astronomy & Astrophysics*.
- **Dimoff** & Orosz (2023). Modelling Apsidal Motion in Eclipsing Binaries using ELC. *The Astronomical Journal*.
- Bergemann, Bensby, Gerber et al. incl. **Dimoff** (in prep). 4MIDABLE-HR: Science motivation and overview of the 4MOST Milky Way disk and bulge high-resolution survey.
- Seeburger, Rix, El-Badry et al. incl. **Dimoff** (2025). The Physical Properties of Post Mass-Transfer Post-Mass-Transfer Binaries. *Astronomy & Astrophysics*.